

Question:

What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?

Answer Choices:

- A. 2.48×10^{-8} cm
- B. 5.92×10^7 cm/sec
- C. 8.66×10^{-6} cm
- D. 3.14×10^{-5} cm
- E. 9.81×10^{-4} cm
- F. 5.27×10^{-9} m
- G. 1.23×10^{-7} cm
- H. 4.14×10^{-10} m
- I. 6.625×10^{-27} erg sec
- J. 0.512×10^6 eV

Reasoning + Answer:

<think>

- Kinetic Energy (KE) = 1 eV = 1.602×10^{-19} J
- Planck's constant (h) = 6.626×10^{-34} Js
- Mass of electron (me) = 9.109×10^{-31} kg
- de Broglie wavelength (λ) = $h / (\text{sqrt}(2 * m * KE))$
- Plugging in the values , $\lambda = (6.626 \times 10^{-34}) / \text{sqrt}(2 * 9.109 \times 10^{-31} * 1.602 \times 10^{-19})$
 $= 1.23 \times 10^{-7}$ cm .

</think>

<answer> 1.23×10^{-7} cm</answer>

